Let A be the event of getting exactly two heads,

B be the event of getting at most two heads, and

C be the event of getting at least two heads.

When ten coins are tossed, the number of elements in sample space is $2^n = 2^{10} = 1024$

$$n(S) = 1024$$

$$n(A) = {}^{10}C_2 = 45$$

$$n(B) = {}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 = 1 + 10 + 45 = 56$$

$$n(C) = {}^{10}C_2 + {}^{10}C_3 + {}^{10}C_4 + \cdots + {}^{10}C_{10}$$

$$= n(S) - ({}^{10}C_0 + {}^{10}C_1) = 1024 - 11 = 1013$$

The required probabilities are

(i) $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{45}{1024}$

(ii) $P(B) = \dfrac{n(B)}{n(S)} = \dfrac{56}{1024} = \dfrac{7}{128}$

(iii) $P(C) = \dfrac{n(C)}{n(S)} = \dfrac{1013}{1024}.$

Example 12.5

Suppose a fair die is rolled. Find the probability of getting

(i) an even number (ii) multiple of three.

Solution

Let S be the sample space,

A be the event of getting an even number,

B be the event of getting multiple of three.

Therefore,

$S = \{1, 2, 3, 4, 5, 6\}$. $\Rightarrow n(S) = 6$

$A = \{2, 4, 6\}$ $\Rightarrow n(A) = 3$

$B = \{3, 6\}$ $\Rightarrow n(B) = 2$

The required probabilities are

(i) P (getting an even number) = $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{3}{6} = \dfrac{1}{2}$

(ii) P (getting multiple of three) = $P(B) = \dfrac{n(B)}{n(S)} = \dfrac{2}{6} = \dfrac{1}{3}.$

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